

# SCIENCE MASTER X

Complete Academic Study Notes • Class 10 Science

## CHAPTER 11

# Electricity

(Full Chapter)

Electric Current & Circuit • Potential Difference • Ohm's Law • Factors Affecting Resistance & Resistivity • Combination of Resistors (Series & Parallel) • Heating Effect of Electric Current • Electric Power & Commercial Energy Units

Target Level: CBSE Class 10

File Name: chapter11x.pdf

## Chapter Overview & Basic Concepts

Electricity plays a central role in modern society. It is a controllable and convenient form of energy used in homes, schools, hospitals, and industries.

### SECTION 1: ELECTRIC CURRENT & ELECTRIC POTENTIAL

#### 1.1 Electric Charge & Electric Current

- **Electric Charge (\$Q\$):** An intrinsic property of matter. SI Unit: **Coulomb (C)**. Charge on 1 electron (\$e\$) =  $1.6 \times 10^{-19} \text{ C}$ .
- **Quantization of Charge:**  $Q = ne$ , where  $n$  is the number of electrons. ( $1 \text{ C} \approx 6.25 \times 10^{18}$  electrons).
- **Electric Current (\$I\$):** The rate of flow of electric charges through a conductor per unit time.
- **Direction of Current:** Conventionally taken as opposite to the direction of flow of electrons (from positive to negative terminal).
- **SI Unit of Current: Ampere (A)** ( $1 \text{ A} = 1 \text{ C/s}$ ). Measured using an **Ammeter** connected in *series* in a circuit.

#### 1.2 Electric Potential & Potential Difference

- **Electric Potential Difference (\$V\$):** Work done to move a unit positive charge from one point to another in an electric circuit.
- **SI Unit of Potential Difference: Volt (V)** ( $1 \text{ V} = 1 \text{ J/C}$ ). Measured using a **Voltmeter** connected in *parallel* across the component.

##### Important Basic Formulae:

- Electric Current:  $I = \frac{Q}{t}$
- Charge Quantization:  $Q = ne$
- Potential Difference:  $V = \frac{W}{Q}$

 Diagram: Schematic Circuit Diagram



**NCERT High-Yield Insight:** An ideal ammeter has **zero resistance** (connected in series), while an ideal voltmeter has **infinite resistance** (connected in parallel) so that it does not alter the circuit current.

SECTION 2: OHM'S LAW & RESISTIVITY

2.1 Ohm's Law

The potential difference ( $V$ ) across the ends of a given metallic wire in an electric circuit is directly proportional to the current ( $I$ ) flowing through it, provided its temperature remains constant.

Ohm's Law Formula:

- $V \propto I \implies V = IR$
- Resistance:  $R = \frac{V}{I}$  (SI Unit: **Ohm,  $\Omega$** )

2.2 Factors Affecting Resistance of a Conductor

- Length of Conductor ( $L$ ):** Resistance is directly proportional to length ( $R \propto L$ ). Doubling length doubles resistance.
- Area of Cross-Section ( $A$ ):** Resistance is inversely proportional to cross-sectional area ( $R \propto \frac{1}{A}$ ). Thicker wires offer less resistance.
- Nature of Material:** Represented by electrical resistivity ( $\rho$ ).
- Temperature:** Resistance of pure metals increases with an increase in temperature.

2.3 Electrical Resistivity ( $\rho$ )

Resistivity is a fundamental property of a material that measures how strongly it resists electric current.

- Formula:**  $R = \rho \frac{L}{A} \implies \rho = \frac{R A}{L}$
- SI Unit: Ohm-meter ( $\Omega \cdot \text{m}$ ).**
- Key Distinction:** Resistivity depends ONLY on the material and temperature, NOT on length or thickness of the wire.

2.4 Material Classification Table

| Category   | Resistivity Range ( $\Omega \cdot \text{m}$ ) | Examples & Applications                                                                      |
|------------|-----------------------------------------------|----------------------------------------------------------------------------------------------|
| Conductors | $10^{-8}$ to $10^{-6}$                        | Silver, Copper, Aluminum (Used for transmission wires).                                      |
| Alloys     | Higher than constituent metals                | Nichrome, Constantan (Used in heating element—does not oxidize easily at high temperatures). |
| Insulators | $10^{12}$ to $10^{17}$                        | Rubber, Glass, Wood (Used for insulation and safety coatings).                               |

SECTION 3: COMBINATION OF RESISTORS

3.1 Resistors in Series

- Resistors joined end-to-end sequentially.
- Current (\$I\$):** Remains SAME through each resistor.
- Voltage (\$V\$):** Divides across resistors (\$V = V\_1 + V\_2 + V\_3\$).
- Equivalent Resistance (\$R\_s\$):**  $R_s = R_1 + R_2 + R_3 + \dots + R_n$
- The total equivalent resistance is greater than the highest individual resistance.

3.2 Resistors in Parallel

- Resistors connected across two common point nodes.
- Voltage (\$V\$):** Remains SAME across each resistor.
- Current (\$I\$):** Divides through branches (\$I = I\_1 + I\_2 + I\_3\$).
- Equivalent Resistance (\$R\_p\$):**  $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$
- The reciprocal of total resistance is the sum of reciprocals of individual resistances. Equivalent resistance is smaller than the smallest individual resistance.

3.3 Comparison Table: Series vs. Parallel Circuits

| Parameter             | Series Combination                            | Parallel Combination                                          |
|-----------------------|-----------------------------------------------|---------------------------------------------------------------|
| Current               | Same through all resistors                    | Divides among branches                                        |
| Potential Difference  | Divides across resistors                      | Same across all branches                                      |
| Equivalent Resistance | Increases ( $R_s = R_1 + R_2$ )               | Decreases ( $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$ ) |
| Circuit Failure       | If one component breaks, entire circuit fails | If one component fails, others work normally                  |
| Domestic Use          | Not suitable for home wiring                  | Ideal for domestic appliances                                 |

Diagram: Series vs Parallel Resistor Layout



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### SECTION 4: HEATING EFFECT & ELECTRIC POWER

#### 4.1 Joule's Law of Heating

When an electric current flows through a purely resistive conductor, electrical energy is dissipated entirely in the form of heat energy.

- **Joule's Law Statement:** Heat produced ( $H$ ) in a resistor is directly proportional to:
  1. Square of current ( $I^2$ ) for a given resistance.
  2. Resistance ( $R$ ) for a given current.
  3. Time ( $t$ ) for which current flows.

#### Heating & Power Formulae:

- Heat Produced:  $H = I^2 R t = V I t = \frac{V^2}{R} t$
- Electric Power:  $P = \frac{W}{t} = V I = I^2 R = \frac{V^2}{R}$  (SI Unit: **Watt, W**)
- Commercial Unit of Energy:  $1 \text{ kWh} = 3.6 \times 10^6 \text{ Joules (J)} = 1 \text{ Unit}$

#### 4.2 Practical Applications of Heating Effect

- **Electric Heating Appliances:** Heaters, irons, and toasters use heating elements made of alloys like **Nichrome** due to high resistivity and high melting point.
- **Electric Bulb Filament:** Made of **Tungsten** (very high melting point =  $3380^\circ\text{C}$ ). Bulbs are filled with inactive gases like *Argon* or *Nitrogen* to prevent oxidation.
- **Electric Fuse:** Safety device connected in *series* with the live wire. Made of a low melting point alloy (e.g., lead and tin). Melts and breaks the circuit when current exceeds safety limit.

### ? Important Practice Questions

**Q1: How much current will an electric bulb draw from a 220 V source, if the resistance of the bulb filament is 1200  $\Omega$ ?**

**Ans:** Given:  $V = 220 \text{ V}$ ,  $R = 1200 \Omega$ .

Using Ohm's law:  $I = \frac{V}{R} = \frac{220}{1200} = 0.183 \text{ A}$ .

**Q2: A wire of given material having length  $L$  and area of cross-section  $A$  has a resistance of 4  $\Omega$ . What would be the resistance of another wire of the same material having length  $L/2$  and area of cross-section  $2A$ ?**

**Ans:** Initial resistance  $R_1 = \rho \frac{L}{A} = 4 \Omega$ .

New resistance  $R_2 = \rho \frac{(L/2)}{(2A)} = \frac{1}{4} \left( \rho \frac{L}{A} \right) = \frac{1}{4} \times 4 \Omega = 1 \Omega$ .

**Q3: Why are copper and aluminum wires usually employed for electricity transmission?**

**Ans:** Copper and aluminum have extremely low electrical resistivities. Therefore, they allow electric current to flow easily without generating excessive heat loss during transmission.

**Q4: An electric heater of resistance  $8\ \Omega$  draws  $15\ \text{A}$  from the service mains for 2 hours. Calculate the rate at which heat is developed in the heater.**

**Ans:** "Rate of heat developed" means Electric Power (P).

Given:  $R = 8\ \Omega$ ,  $I = 15\ \text{A}$ .

$$P = I^2 R = (15)^2 \times 8 = 225 \times 8 = \mathbf{1800\ \text{W (or 1800 J/s)}}.$$